

In a shipyard, a loaded transporter does not drive on the road. It owns the road.

Network-bound capacity, deadlock-free orchestration and fleet-network co-design
(working title)

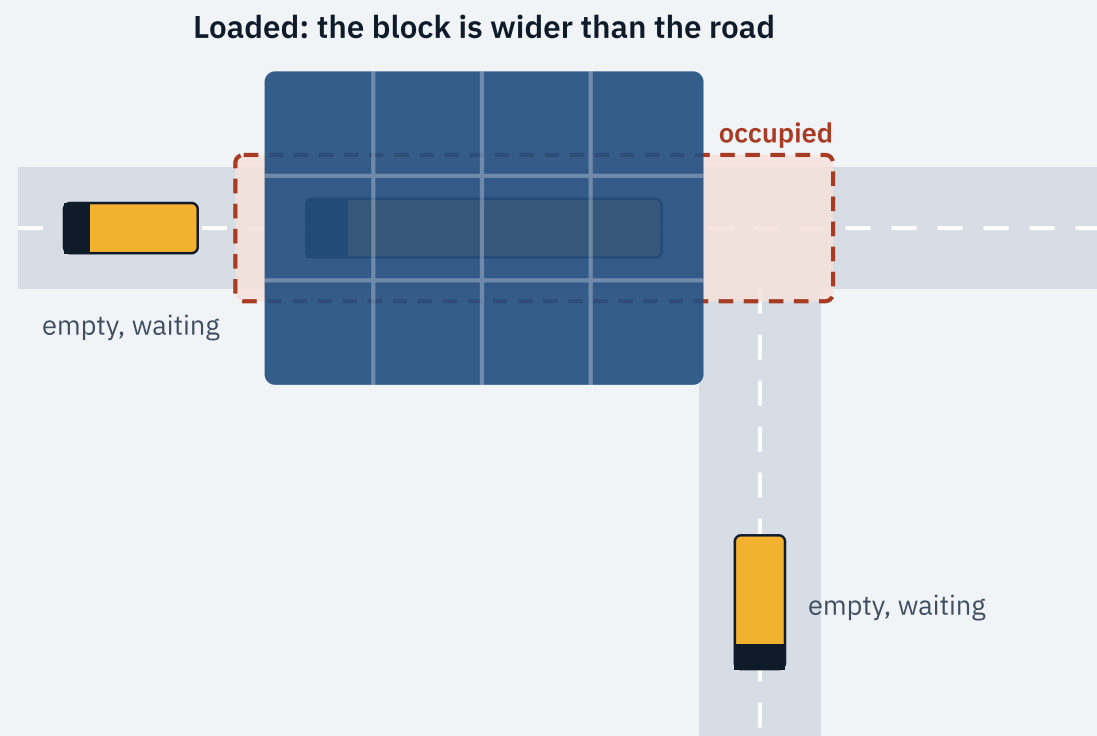
Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

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Direction set 8 October 2026 · flag experiment done on two yards

Free-flow model: vehicles never meet

Occupancy model: loaded vehicles own the road



The whole thesis in one minute

1

The premise

Shipyard transport scheduling almost always assumes a free-flow network: travel times from a distance matrix, vehicles that never interact.

2

The question

When loaded vehicles own the road, what sets capacity, and how should fleet, scheduling and network be designed together?

3

First results

Both yards reach a throughput plateau set by the network, not the fleet. On Okpo, from 15 to 30 vehicles each extra unit adds only 58%→34% of its free-flow gain; the dock mouth is a local bottleneck at erection peaks.

4

Next

Prove T1, T3, T6 (two bounds, unbounded free-flow error, road–vehicle exchange rate); submit the first paper within 3–4 months; extend to 8, then 15 yards.

Where the network binds, fix the road rather than buy a vehicle, and schedule the road rather than the vehicles.

The order of this briefing (click to jump)

1

Background and premise

Blocks wider than roads; free flow by default; a gap named by peers p. 4

3

Model and theory

Blocking-time model, theorems T1–T6 p. 10

5

Flag experiment

Okpo and Yantai: fleet size against throughput, the dock-mouth bottleneck p. 14

7

Progress and plan

Eight-week plan, objections and answers, the results ladder p. 20

2

Question, claims, hypotheses

The core question, five claims, seven two-sided hypotheses p. 7

4

Thesis plan and methods

Seven chapters, four journal papers, algorithms p. 12

6

Data and many yards

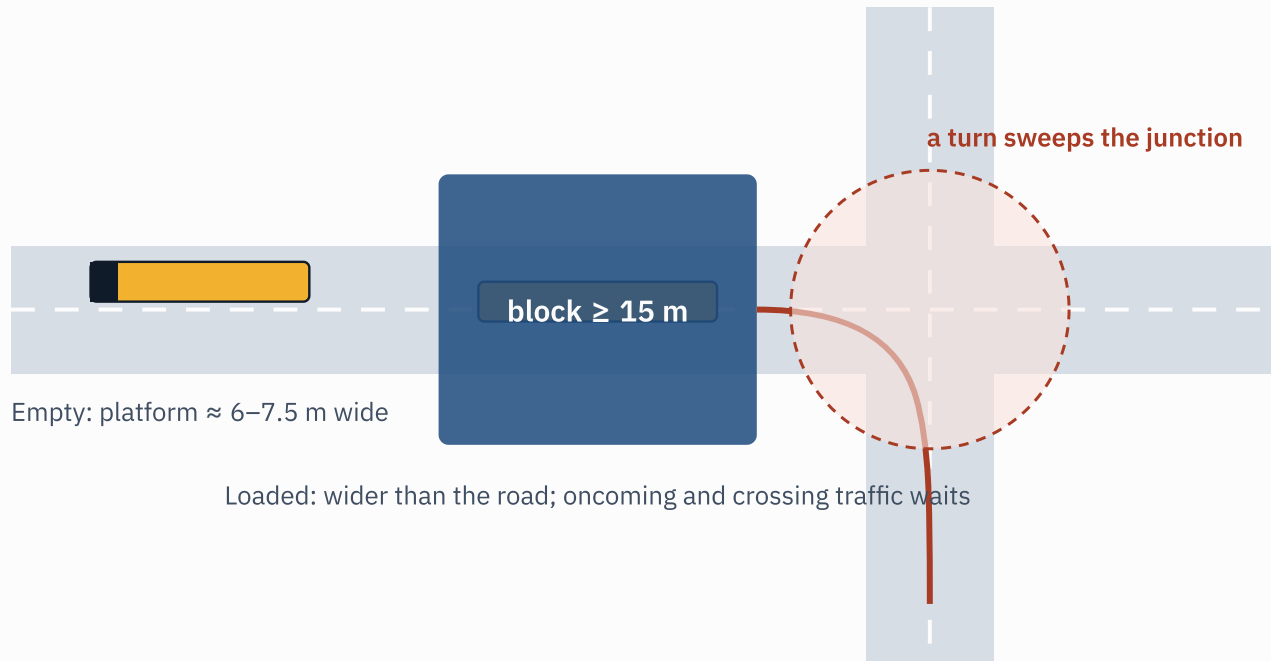
Public data only; a list of 15 yards p. 19

8

Earlier work

How earlier results serve the thesis p. 23

Blocks are far wider than their carriers: once loaded, a transporter holds the whole segment and junction



- **Empty platforms** are about 6–7.5 m wide; one Hanwha Ocean transporter is 21 m × 7.5 m and drives with signallers ahead and behind
- **Loaded outlines** are often above 15 m, e.g. blocks of about 15 × 15 × 12 m
- **The largest carriers** have platforms of about 30 m × 10 m and occupy the road even when empty

The gap between loaded and empty, and between large and small carriers, is particular to shipyards and weak for AGVs and warehouse robots.

The field's default: a free-flow network in which vehicles never interact

TYPE	WORK	TREATMENT	DISTANCE TO OCCUPANCY
Excluded	Meng et al. 2018	Path overlap in the stockyard only; road interference explicitly excluded	free flow
Free flow	Liu et al. 2022; Zhou et al. 2025; Han et al. 2024	Distance matrices or departure and arrival events only; some recommend fleet sizes on that basis	free flow
Turning delay	Moon et al. 2017	Per-vehicle penalties for turning at junctions, changing each vehicle's shortest path	not occupancy between vehicles
Soft penalty	Roh & Cha 2011; Heo et al. 2013	Interference counted in the objective; penalty when two blocks share a segment at the same time	not a structural capacity constraint; loaded and empty not distinguished; no fleet or network design
Single-yard simulation	Tao et al. 2017	Road capacity and “road interference” in a simulation; utilisation rises after reconfiguring vehicles	mechanism unclear; one yard

No one has treated the exclusive use of segments and junctions by loaded vehicles as a structural capacity constraint, or used it to size fleets or upgrade networks (within our search).

Sources: direction document §1.1, checked against the originals on 9 Oct 2026. Full texts: Meng 2018 (line 302), Liu 2022, Zhou 2025 (assumes “traffic congestion not considered”), Roh & Cha 2011 (eqs. 2, 7), Heo 2013, Moon 2017; Han 2024 is a two-page poster, Tao 2017 abstract only. Search scope: two searches of 8 Oct 2026 and the literature library (shipyard scheduling, multi-agent path finding, railway capacity, queueing networks with blocking).

Peers have named the gap, and the yard data show it

A stated gap: Sun, Liang, Chang (2024), *PLOS ONE*

“how to address the interference limitations of roads in the yard remains a limitation of this study. In actual transportation operations, some roads can only accommodate one transporter at a time.”

Chang Daofang's group, Shanghai Maritime University; they may well follow up, so speed matters.

Yard evidence: Wang, Yin, Khan (2024), *JMSE*

288 delays at a Shanghai yard in 2022: transporters take several lanes and other vehicles must give way; each road-environment event (other vehicles, obstacles, pedestrians) costs about 15 min.

Other fields: more vehicles jam, road changes move capacity

- **Container-terminal AGV simulation:** going from 60 to 80 vehicles lowered hourly throughput (by about 2%) Cheng et al. 2005
- **Terminal traffic network:** saturation near 100 vehicles; without keep-clear hatching at junctions, deadlocks from 30 Zhou et al. 2017
- **Underground mine:** single-width roadways with passing bays, trucks managed on railway block-section logic Eustace et al. 2021

These are single-site simulations or operating observations; the shipyard's loaded-to-empty footprint ratio is the most extreme of them.

The core question: when loaded vehicles own the road, what sets capacity?

When loaded vehicles occupy the road exclusively, what determines the capacity of a shipyard block-transport system, and how should fleet, scheduling and network be designed together?

The usual view

- Transport capacity = fleet capacity
- Scheduling means assigning jobs to vehicles
- More vehicles are better, with diminishing returns at worst

The road-occupancy view

- Loaded vehicles are moving road owners; capacity = $\min(\text{fleet}, \text{network})$
- Where the network binds, schedule the road rather than the vehicles
- Fix a road rather than buy a vehicle; more vehicles can even slow the system

Five central claims in one figure

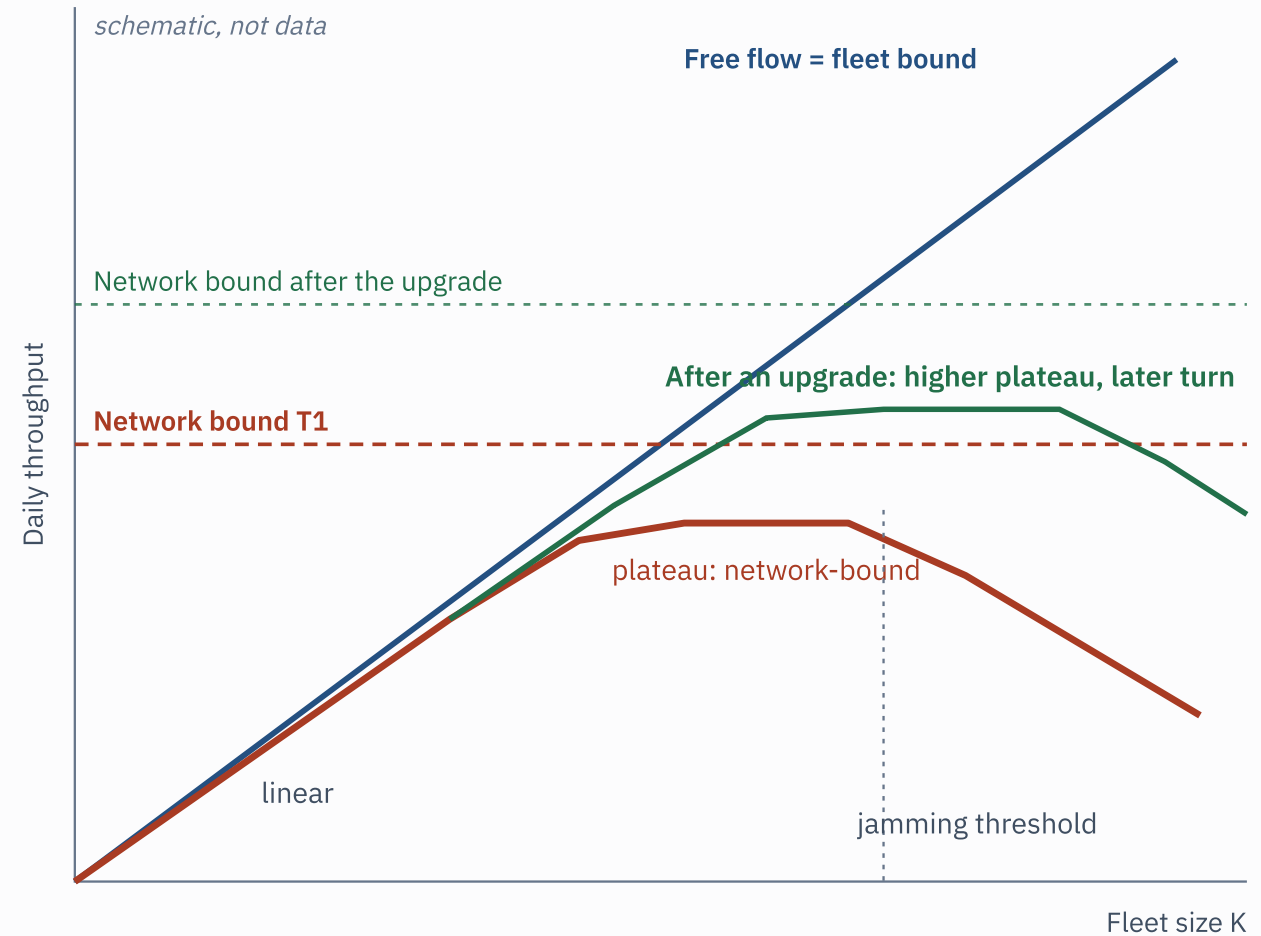
- C1 Two upper bounds**
Throughput $\leq \min(\text{fleet bound}, \text{network bound})$; the network bound is an exact linear program

- C2 Fleet-throughput trapezoid**
Linear rise, then a network plateau, then decline past a jamming threshold; free flow sees only the first part

- C3 Dispatching becomes road scheduling**
Where the network binds, value comes from release order, waiting points and segment reservations; naive dispatching deadlocks

- C4 Road-vehicle exchange rate**
Passing bays, one-way loops, holding areas and time windows can replace vehicles, at the network constraint's dual price in vehicles

- C5 Transferable**
Jamming thresholds and upgrade values can be predicted across yards from a few topological indicators



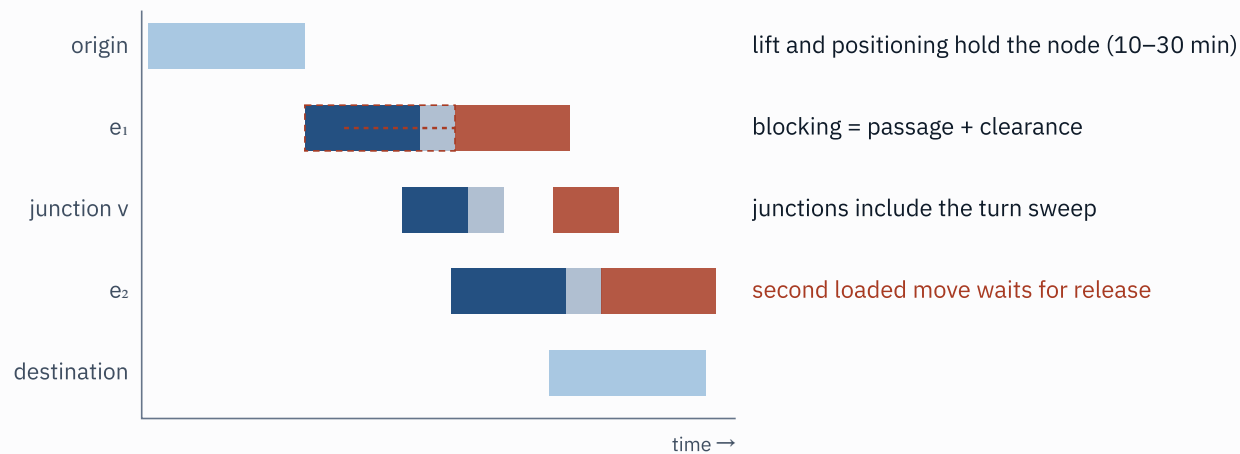
■ Free flow = fleet bound ■ Occupancy model ■ After an upgrade: higher plateau, later turn

Seven hypotheses fixed in advance, each conclusive either way

#	HYPOTHESIS	IF IT HOLDS	IF IT FAILS
H1	At realistic workloads and usual fleet sizes, the system is network-bound, at least at assembly peaks or dock mouths	Network-bound capacity exists	A density criterion for when free flow suffices
H2	Under common dispatching, throughput is a trapezoid or inverted U in fleet size; good orchestration delays the turn but cannot remove it	A jamming threshold exists	A network-redundancy indicator
H3	Free flow's underestimate of completion time and vehicles grows superlinearly with density	Free flow is systematically wrong at peaks	A usable correction factor
H4	One passing bay or one-way change on a critical link is worth at least one vehicle; dual prices find the link	The exchange rate holds	The bottleneck is dispatching; chapter 4 is strengthened
H5	Naive dispatching deadlocks at peaks; a provably safe reservation policy loses little throughput	Safe orchestration is cheap	A cost formula for safety and its conditions
H6	Thresholds and upgrade values are predictable across yards from a few topological indicators	A transferable empirical law	What transfers is the C1 linear-programming tool
H7	Vehicle type interacts with occupancy: large units occupy loaded and empty; small ones yield when empty but run extra empty trips to team up	The best fleet mix shifts with network constraint	Vehicle type and network can be decided separately

Source: direction document §3. H1 has been restated piecewise from the two yards' readings (decided 8 Oct 2026; see part 5).

The model: segments and junctions are exclusive resources held for a blocking time



- **Road graph** $G = (V, E)$: segments e and junctions v are exclusive resources
- **A loaded move** holds each resource on its path for a blocking time, as in railway signalling
- **Empty runs by class**: large units occupy even when empty; small ones can pass on passable links
- **No road widths measured**: passability classes (loaded-exclusive / passable by empties / two-way) are scenario parameters; results are functions of the passable share

One move holds: $\tau_r = \text{pass}_r + \text{clear}_r$, $r \in \text{path}$

■ loaded move A ■ clearance ■ loaded move B ■ lift, positioning

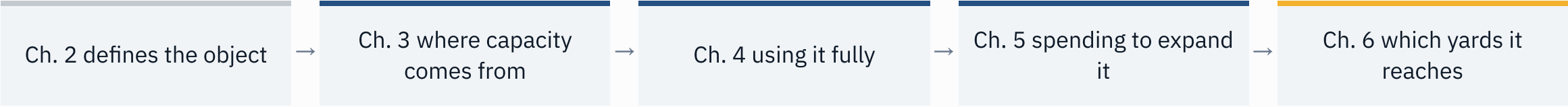
Theory: six theorems, each provable by one person

THEOREM	STATEMENT	BUILDS ON	EFFORT
T1 two bounds	Any stable policy's throughput $\leq \min\{\text{fleet bound, network bound}\}$; the network bound is a multicommodity-flow LP with every resource's utilisation ≤ 1	UIC 406; Goverde 2007	easy, 1–2 wk
T2 trapezoid	Closed-form piecewise throughput in K on corridors and loops, with the jamming threshold; new: asymmetric loaded and empty occupancy	Farhi et al. 2016; Dallery & Towsley 1991	medium, 1–2 mo
T3 unbounded error	A construction in which the ratio of true to free-flow completion time is arbitrarily large (e.g. all blocks share one dock-mouth lane)	construction	easy, 1 wk
T4 non-monotone	Under greedy or first-come dispatching, throughput on a one-bay corridor strictly falls beyond some K_0	sample paths; Onvural 1990	medium, 1 mo
T5 safe orchestration	“Reserve to the next safe holding point before departing loaded” is deadlock-free, with a loss bound relative to T1	Reveliotis 2000; Kim et al. 2006	medium, 1 mo
T6 exchange rate	From the dual of the T1 LP: upgrading link e is worth its dual price, converted to vehicles; only upgrades on tight constraints pay	LP duality	easy, 1–2 wk
Complexity	Feasibility with deadlines and exclusive resources is NP-hard	Mascis & Pacciarelli 2002	easy

T1, T3 and T6 first (highlighted): together they carry the theory of the first paper; T2, T4 and T5 add to it. Only graph theory, linear programming, max-plus algebra, sample paths and constructive counterexamples; no heavy stochastic asymptotics.

Thesis plan: seven chapters and four journal papers answering one question

CH.	TITLE	QUESTION	OUTCOME	TARGET
1	Introduction	Where free flow comes from and why it fails in yards	A literature map across four fields	—
2	Occupancy model	How to represent wider loads, lifting at nodes, swept junctions	Blocking-time graph; complexity; public-data instance generator	with ch. 3
3	Capacity	Where capacity comes from, when the network binds, when vehicles hurt	T1–T4	TR-B / TS
4	Orchestration	How to assign jobs and order passages without deadlock	Exact model, large-scale search, T5, rolling online	TR-C / T-ASE / EJOR
5	Fleet–network co-design	Buy vehicles or fix roads; large or small vehicles	T6; critical links; vehicle-type interaction (H7); bilevel algorithm	TR-E / IISE T
6	Many yards	Do the laws generalise and predict?	8–15 public layouts; topological prediction (H6)	OE / C&IE
7	Conclusion	—	—	—



Source: direction document §4 and §8.2. TR-B/C/E = Transportation Research Part B/C/E; TS = Transportation Science; T-ASE = IEEE Trans. Automation Science and Engineering; IISE T = IISE Transactions; OE = Ocean Engineering; C&IE = Computers & Industrial Engineering.

Algorithms, and what the thesis brings back to three research communities

Algorithms (chapters 4–5)

- **Exact model:** CP-SAT interval variables, one no-overlap constraint per segment and junction; ground truth on small instances
- **Large scale:** integrated assignment-and-sequencing neighbourhood search; priority-based planning for conflicts; compared with “schedule in free flow, then repair conflicts”
- **Safety layer:** the T5 reservation policy as a shell around any heuristic
- **Rolling online:** re-plan under uncertain completion times; release with forecasts
- **Co-design:** outer level picks upgrades and fleet, inner level evaluates by orchestration; T6 dual prices rank the outer options
- **Optional frontier:** LLM-evolved orchestration rules (EoH, ReEvo, FunSearch; no shipyard use found)

Framing: “we extend X to Y and find that Z changes under Y”

Multi-agent path finding

A new agent class: footprint multiplies when loaded, long lifts at endpoints, fleet size to be chosen; existing large agents have fixed shapes and no tasks

Railway capacity

Blocking times and compression carried to trackless road networks with asymmetric loaded and empty occupancy

Queueing networks with blocking

A real system that breaks the vehicle–hole symmetry, because empty and loaded vehicles occupy differently

The flag experiment: two real yard layouts, three traffic rules, one task stream

- Free flow

No occupancy check; vehicles never interact (the field's default)
- Whole-route reservation

Reserve the whole route before departing loaded, leave just in time; cannot deadlock
- Segment-by-segment request

Request each segment in turn; deadlocks cleared by hand after 900 s (no published value, reference only)
- **Three demand levels:** regular, peak (twice regular), saturated (tasks never run out, so it measures capacity)
 - **Fleet size** $K = 5\text{--}150$ (Yantai from 1), 10 seeds per point, 16-h days
 - **Vehicles:** 12 km/h empty, 6 km/h loaded, 10 min to load and to unload; platform 16.25×5.5 m; blocks 5–15 m wide, up to 30 m long
 - **Bounds:** the T1 LP; T1' adds two width bin-packing rows per resource (Fekete–Schepers dual-feasible functions, exact rationals), valid for both occupancy rules, 14% tighter on Okpo

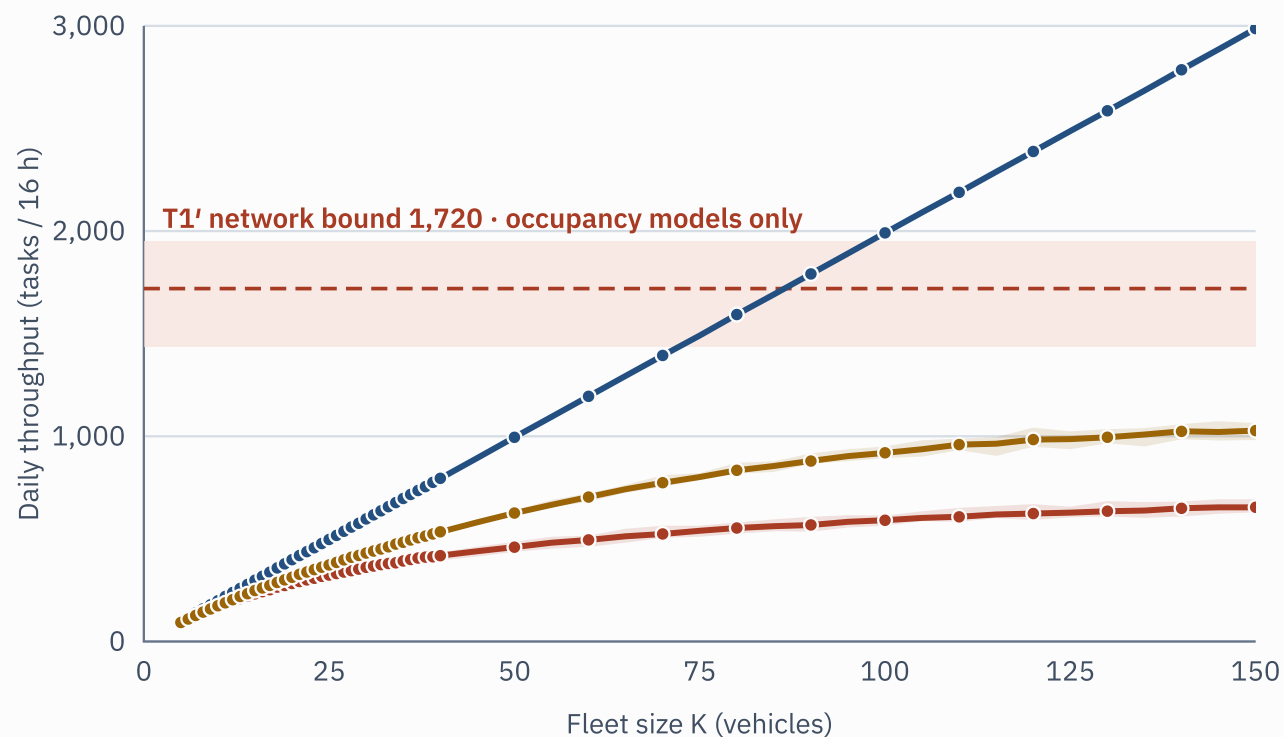
	OKPO	YANTAI
Roads (count / km)	721 / 40.3	140 / 8.6
Junction resources	334	58
Entrances	218	32
Stops	421	140
Independent cycles	82	14
Tasks a day: regular / peak	97 / 194	23 / 46

Okpo's workload follows MY-B (Roh & Cha 2011); Yantai's 23 tasks a day come from its public steel use of 68,512 t a year (100 t blocks, 10 moves each, 300 working days).

Sources: thesis-core commit e7a2732 (T17–T20, each reviewed by the project control and an independent reviewer); speeds from the Suzhou Dafang DCY270 data sheet; T1' proof in docs/T18_T1_tight_proof.md. Networks were traced by the author on public imagery; neither imagery nor network geometry is shown.

The flag figure: free flow keeps rising; the occupancy models reach a plateau that more vehicles cannot lift

Okpo Yantai **Saturated (capacity)** Peak



Free flow Whole-route reservation Segment request

T1' network bound (occupancy models only)

628 / 984

Saturated plateau: reservation / segment (a day, mean of K = 100–150)

1,720

Tightened network bound T1' (a day; T1 = 1,997 before tightening)

36%

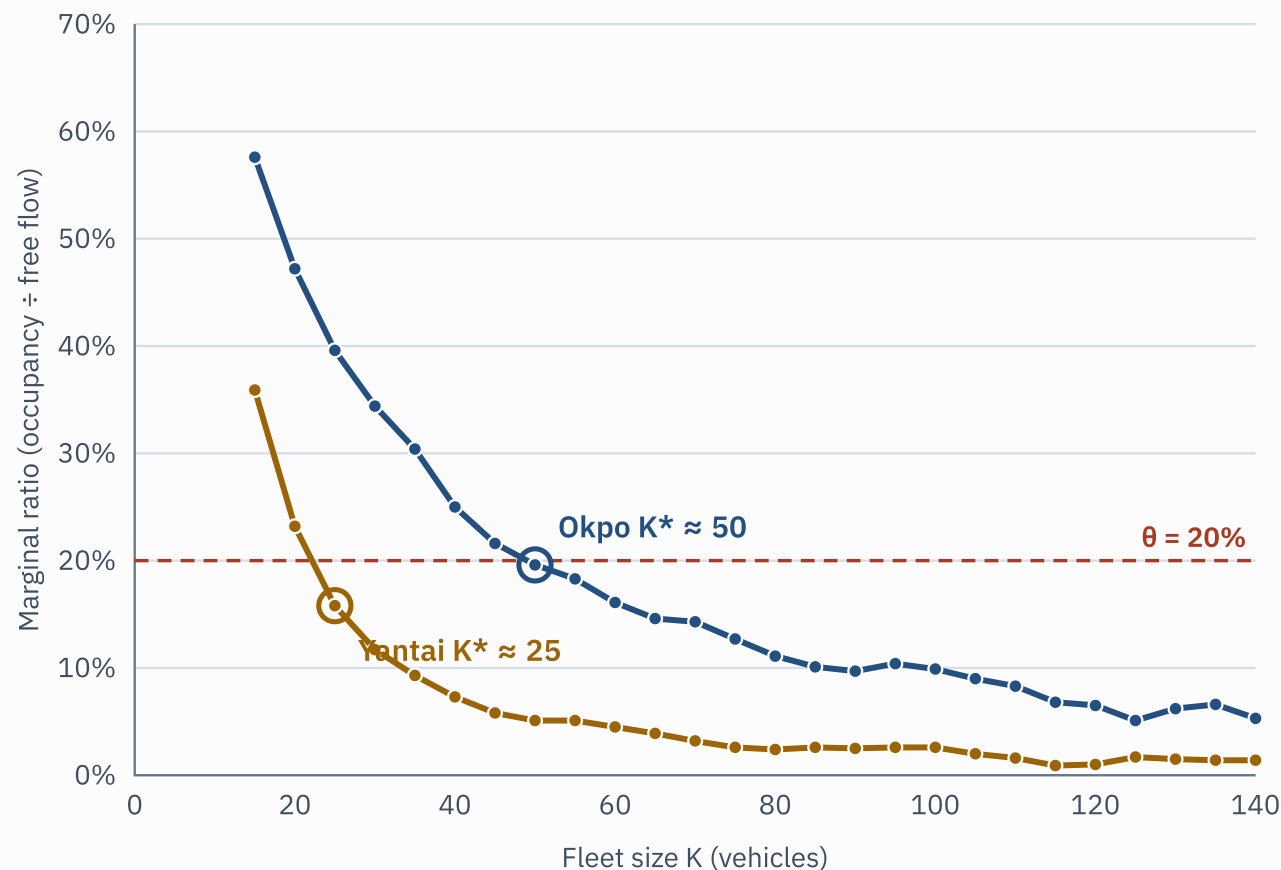
The reservation plateau reaches only this share of T1': the rest is the scheduling rule, which chapter 4 aims to win back

K ≈ 50

Network-bound from here (marginal ratio first below 20%; range 50–60)

Sources: thesis-core e7a2732; Okpo results/a0ab16fceb94 and 49d2383872e7, Yantai results/b82aa19b22af (map rev 395). Means of 10 seeds, shading min–max. Throughput = transport tasks completed per 16-h day (one move each). T1 and T1' bound only the occupancy models; free flow does not check capacity and may exceed them.

Reading H1: the usual fleet sizes already feel the network, and where it takes over depends on the layout



■ Okpo ■ Yantai — $\theta = 20\%$

$K \approx 50$

Okpo: network-bound from here
(range 50–60)

$K \approx 25$

Yantai: much earlier
(same under both OD samplings)

58% → 34%

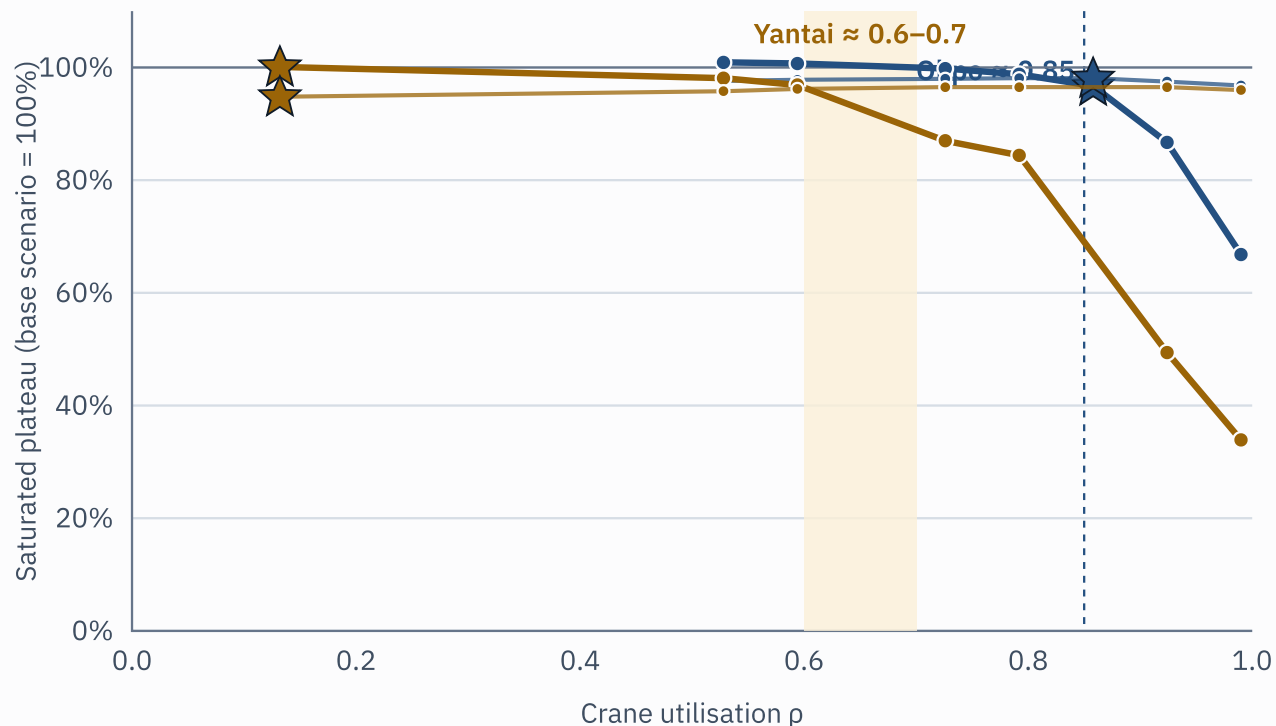
Okpo, $K = 15 \rightarrow 30$: each extra vehicle's gain as a share of free flow

10 → 12

Okpo vehicles needed at peak: free flow → occupancy

H1, restated from the readings: at saturated demand there is a throughput plateau set by the network and independent of fleet size, starting at a layout-dependent K. At the usual 10–30 vehicles, occupancy already changes capacity and fleet sizing.

The dock mouth is a local bottleneck at erection peaks: past a crane-utilisation threshold, vehicles waiting for the crane block the dock road and the yard's plateau falls steeply



- **Dock mouth = crane and dock stopping road in series.** Crane utilisation $p = \text{erections per day} \times 190 \text{ min} \div (\text{cranes} \times 960 \text{ min})$
- **Segment request:** Okpo's plateau holds for $p \leq 0.79$, -3% at 0.86 (the current working point), -13% at 0.92, -33% at 0.99 with no steady state; Yantai drops earlier, -13% already at 0.73
- **Whole-route reservation:** vehicles wait at their last unloading point, so the plateau does not move with p ; the cost moves to erection flow time and peak fleet
- **The LP cannot see it:** the dock margin stays near 1.46 at every p ; this is queue-induced time blocking that only the occupancy simulation reveals

■ Okpo · Segment request ■ Yantai · Segment request --- Whole-route reservation

★ current working point

Sources: thesis-core e7a2732, t20_summary.json (8-day runs; plateau = mean over $K = 100-150$, relative to each yard's base scenario). Cranes: every dock crane of ≥ 100 t counts, 3 per yard; hook 10 min and positioning 180 min from ZhangSG2025 (another yard). Working points: Okpo 13 erections a day ($p = 0.86$), Yantai 2 ($p = 0.13$). Delivery by transporter to the dock edge is a scenario assumption; in-span direct lifts are being classified in T21.

Two yards: a plateau everywhere, a bottleneck that moves

	OKPO	YANTAI
Roads (count / km)	721 / 40.3 km	140 / 8.6 km
Plateau, whole-route reservation	628	455
Plateau, segment request	984	628
Network bound T1'	1,720	1,252
Reservation plateau ÷ T1'	36%	36%
Network-bound from K*	≈ 50	≈ 25
Vehicles at peak (free flow → reservation)	10 → 12	2 → 2
Where T1' binds	a dock stopping road	the door of a single-door shop
Crane working point p (erections a day)	0.86 (13)	0.13 (2)

Common to both

- A plateau independent of fleet size (H1; the first two parts of C2)
- The reservation plateau is about 36% of T1' on both: the gap between plateau and bound comes from the scheduling rule, and is what chapter 4's orchestration aims to win back

Yard-specific

- Where it binds: Okpo at a dock stopping road, Yantai at the door of a single-door shop
- The start K* and the dock's p threshold differ; on Yantai, letting empties pass is the largest single lever (+58% at steady state)

Sources: thesis-core e7a2732 (T18–T20). How close Yantai's plateau sits to its bound also depends on OD sampling (T19); at its public-output demand Yantai needs only 1–2 vehicles, so occupancy does not change its fleet. Which quantities topology can predict is what C5 and H6 test on more yards.

Public data only, no road widths measured; chapter 6 tests 15 yards

NEED	SOURCE
Layouts	Centrelines and junctions traced on public imagery, topology only; OSM roads as drafts where available
Passability	Scenario parameter, three classes scanned by share
Task flows	Published data and public output figures; synthetic parts labelled as scenarios
Vehicles	Manufacturers' public specifications
Anchor	Wang 2024: about 15 min per road-environment event

Not needed: measured widths, GPS traces, company data. Theorems need no data; numerical results are stated for the listed layouts and scenarios.

Done: 2

Hanwha Ocean Okpo (Korea) · CIMC Raffles Yantai (China)

Group A, 6: eight yards in six countries

Chantiers de l'Atlantique (FR) · Meyer Werft Papenburg (DE, covered dock halls)
· Imabari Saijo (JP) · Philly Shipyard (US) · New Yangzi (CN) · Samsung Heavy
Geoje (KR)

Group B, 7: up to 15

Fincantieri Monfalcone · Oshima · Namura Imari · CM Jinling Weihai · Hengli · HD
Hyundai Samho · HD Hyundai Mipo

Selection: civilian yards; open data first (open imagery, OSM roads, public filings); wide variety of size, layout, dock type, region and product, so the topological prediction of chapter 6 is a strong test.

Sources: direction document §7; yard list and open imagery sources from the project's yard list (9 Oct 2026). Thesis figures show only our traced vector networks, never satellite imagery.

Progress: two days after the direction was set, the flag curve exists for two yards

WHEN	ITEM	OUTPUT	STATUS
10-08	Direction set: road-occupying transport	Title, five claims, seven hypotheses, theorem list, eight-week plan	done
10-08	Flag experiment (Okpo): blocking-time model, simulation of three rules, T1 bound	First fleet-size–throughput curve	done
10-08	T1 tightened to T1' with proof; just-in-time departure; H1 readings	Proof; piecewise H1	done
10-09	Second yard, Yantai; erection at crane cadence and the crane-utilisation curve	Two-yard comparison; dock-mouth mechanism figure	done
10-09	Gantry-crane spans and floating docks drawn from imagery; erections split into in-span and out-of-span; crane counts per dock list	Okpo's new crane working point	running
wk 2–5	Prove T1, T3, T6; start T2	Complete proofs of three theorems	author
wk 3–6	Annotating group A, six yards (OSM drafts for seven yards are on open base maps)	Multi-yard instance library	running
wk 4–8	CP-SAT exact model and neighbourhood search; T5 safe policy	Chapter 4 prototype	planned
wk 6–8	First paper (chapter 3): bounds, trapezoid, free-flow error, exchange rate; preprint in parallel	TR-B / TS	planned

Sources: direction document §12 (eight-week plan); task records T17–T21 (8–9 Oct 2026). On pace: Sun 2024 lists single-transporter roads as future work, so the first paper should be submitted within 3–4 months.

The likeliest objections, and the answers

OBJECTION	ANSWER
Yard conflicts are done already Roh & Cha 2011; Heo 2013; Moon 2017	They treat crossings as soft penalties in path search, or turns as time added to one vehicle; none writes exclusivity as a structural capacity constraint, proves capacity results or designs fleet and network. Moon 2017 itself notes earlier work ignored junction delays.
MAPF already has large agents Li et al. 2019	Their shapes are fixed and they carry no pickup-and-delivery tasks, or agents are points and discs; none combines a load-dependent footprint, long lifts and fleet sizing.
Ports and mines know more vehicles jam	Those are single-site simulation observations; here come capacity theorems, an exchange rate and cross-yard prediction. The shipyard's loaded-to-empty footprint ratio is the most extreme of these settings.
Yards have too few vehicles to jam	That is what H1 tests; the two yards already show occupancy changing capacity and fleet size at 10–30 vehicles, with the dock mouth a local bottleneck at erection peaks. Even without jams, a density criterion results.
No measured data	The theorems need no data; numbers are anchored to Wang 2024's delay magnitude; many public layouts; the passable share is scanned over its whole range.
Exclusivity is too strong	Exclusivity is a parameter, not an assumption: results are functions of the passable share, with full exclusivity at one end.

Sources: direction document §10; the 10–30-vehicle answer rests on the flag readings of part 5.

The results ladder: every chapter's floor is a paper on its own; the ceiling aims at top journals

	FLOOR (DONE ONCE DONE)	TARGET	CEILING
3 Capacity	Two bounds and the trapezoid on one yard two yards in hand	T1–T4 proved	Computable bounds on jamming thresholds for general networks
4 Orchestration	Exact model and large-scale search beating “free-flow schedule, then repair”	T5 safe policy with small loss	LLM-evolved rules beating hand-written ones and transferring across yards
5 Co-design	Grid evaluation of fleets and upgrades	T6 exchange rate and critical links	Bilevel co-design with optimality gaps
6 Many yards	8 yards	15+ yards with predictive features	A simple formula for jamming thresholds from topology

Earlier work serves as tools; the thesis starts from a new premise

ASSET	USE IN THE THESIS
Fleet sizing (MY-B)	ALNS and CP-SAT code start chapter 4; price data feed chapter 5's vehicle types and costs
Coupled-carriage ideas	Chapter 5, H7: large carriers against teams under occupancy
Centreline networks and the tracing tool	Production tool for chapter 6's layouts
Remote-sensing segmentation (MY-A, published)	Optional: faster layout extraction for many yards

Open the fleet-sizing (MY-B) briefing →

Open the segmentation (MY-A) briefing →

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

Thank you.

Source: direction document §9 (use of earlier assets). The thesis starts from the premise of §1; earlier work is used as tools and cited.

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